

HUMAN HEALTH

ENVIRONMENTAL HEALTH

GREAT RESULTS BEGIN WITH GOOD PREPARATION



**Titan MPS Microwave
Sample Preparation System**


PerkinElmer
For the Better

THE FIRST STEP TO BETTER ANALYSIS

Maybe you're an environmental, food, or agriculture scientist helping determine the nutritional content or safeguard the authenticity of the food we eat. Or a drug researcher, analyzing impurities that can affect drug efficacy and metabolism. Or an engineer helping develop the eco-friendly technologies and alternative energy sources of the future. Scientists in these industries and more depend on atomic spectroscopy techniques to generate the results and data they need to help them make informed scientific decisions.

And those techniques often depend on microwave sample preparation systems that can deliver complete, uncontaminated samples — each and every time. Bad preparation? Bad results. It's as simple as that.

With the Titan MPS™ microwave sample preparation system, we're delivering on the promise of a simple, safe, cost-effective microwave — one that provides the quick return on investment that labs are looking for in these times of constrained budgets. And it's backed by PerkinElmer, the one-stop, go-to resource for all things atomic spectroscopy. With a team of support individuals who understand, in a deep-seated way, the science behind the goals your lab is trying to achieve.

The Titan MPS microwave sample preparation system.
Yes, it's a microwave. But we think it's a whole lot more.

A yellow tractor is shown from a low angle, plowing a field. The tractor is in the background, slightly out of focus, with a driver visible in the cab. It is pulling a moldboard plow that has turned over a large amount of dark, rich soil. The foreground is dominated by the freshly plowed earth, showing deep furrows and clumps of soil. A green rectangular text box is superimposed over the middle of the image, containing white text. The sky is clear and blue, and a line of trees is visible in the far distance.

PREPARE TO
SEE THE
DIFFERENCE

THIS IS SAMPLE PREP THE WAY IT SHOULD BE



TAKE IT FROM THE TOP

Before you prepare your first sample, you'll notice the difference: The Titan MPS is a top-loading microwave sample preparation system protected by hardware interlocks to ensure safety during operation. And it's simple to load and unload samples, which keeps everyone more productive – and saves your lab time and money.

Under the lid is a system built for rugged, routine laboratory use, with a cylindrical, corrosion-resistant, PFA-coated oven cavity that ensures even microwave distribution. Its gas collection system captures and removes any gases that vent during digestion. The chamber continually evacuates during oven operation, too, keeping the enclosure free of gas or acidic vapors.

And for flexibility, the system supports interchangeable turntables, which enables you to change from one vessel type to another quickly and easily, expanding your working pressure range.

HOLD EVERYTHING

Strong and durable, the vessels themselves are made of tough TFM™ fluoropolymer, which is resistant to harsh reaction

environments, and they slip into the turntable individually, without the need for a heavy rotor – easy in, easy out.

Digestion performance is enhanced further thanks to our unique Direct Pressure Control™ (DPC) and Direct Temperature Control™ (DTC) optical-reaction monitoring systems, which measure each sample vessel individually. Measurement takes place every time a vessel passes the sensor, resulting in a constant adjustment of power level. Complicated connections between vessel and oven are totally eliminated – and with no thermocouples or probes to insert, you get simple, dependable, contamination-free monitoring.

What's more, vessels are serialized and come with a one-year guarantee, for a considerable savings on your consumables budget. And speaking of consumables, we're a leading provider of scientific instruments, services, informatics, and consumables to a variety of industries – so rest assured we can take good care of your lab, from sample preparation to analysis to communication of test results.

SIMPLE, INTUITIVE OPERATION MAKES EVERYONE AN EXPERT

Controlling your Titan MPS microwave sample preparation system couldn't be simpler. An integrated color touch-screen controller, driven by a PowerPC™ CPU and the open source Linux operating system, delivers user-friendly tools for reaction control, data analysis, and storage of digestion data and user methods. The software comes preinstalled with a number of sample methods covering a wide range of digestion conditions – and the system accommodates any of the additional methods you create and come to rely on.

Built to network, the Titan MPS system can be accessed and monitored via a simple, intuitive GUI from any PC on your lab's network – no special software is required.

TITAN MPS SAMPLE PREPARATION SYSTEM OVERVIEW

Includes:

Digestion vessels

Turntable and gas manifold

Direct Temperature Control (DTC)

Direct Pressure Control (DPC)

Startup consumables for approximately 500 runs

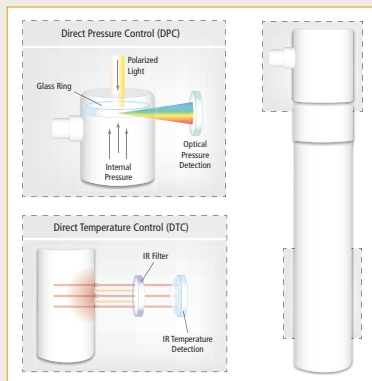
Power cord (US)

Power	230 V, 50 – 60 Hz
Pressure rating	100 bar (8 position) 40 bar (16 position)
Temperature control	Optical/sample temperature
Pressure control	IR/direct sample
Part numbers	N3130100 (16 Position) N3130110 (8 Position)

THE TITAN MPS SYSTEM: WHAT SETS US APART?

Some think all microwave sample preparation systems are the same. But there are some little differences that end up making a big difference:

- This top-loading system with an interlocked lid allows easy loading and removal of vessels
- The cylindrical, pressure-resistant, PFA-coated stainless steel oven chamber and optical temperature and pressure reaction monitoring systems ensure the most reproducible results
- Vessels are reusable and guaranteed for one year, for considerable savings on consumables
- Integrated gas collection system efficiently vents the oven chamber to keep it free of gas or acidic vapors.
- Color touch-screen controller simplifies programming, data storage, and reaction control.





IT'S NOT JUST
THE MICROWAVE
IT'S EVERYTHING
BEHIND IT

ORDINARY MICROWAVES LEAVE YOU A LOT TO DIGEST

When unforeseen problems with your lab's microwave sample preparation system start affecting your results, your first call may be to your microwave manufacturer's service and support representative. But if the cause of the problem isn't easily identified, you'll need to schedule visits from suppliers of other system components – with no guarantee of a quick solution.

But with PerkinElmer, your entire lab becomes our priority. So you can concentrate on what matters most – your science – while we help keep everything up, running, and productive.

WE KNOW YOUR LAB – FROM SAMPLE PREP TO FINAL RESULTS

At PerkinElmer, we're leading providers (and supporters) of the AA, ICP-OES, and ICP-MS instrumentation and data systems at the heart of your lab. So when we're called on to service your Titan MPS microwave sample preparation system, we'll deploy someone who is expert in your complete atomic spectroscopy solution.

Matter of fact, we field more than 1,600 trained and certified service and support professionals around the world, dedicated to helping you realize the maximum productivity and ROI from all the instruments and systems in your lab – including your Titan MPS microwave sample preparation system.

What's more, OneSource® Laboratory Services comprises the most comprehensive portfolio of professional laboratory support offerings in the business, including complete care programs for nearly any technology and manufacturer. And by allowing you to consolidate all your service contracts under one supplier – and by delivering expert technical advice and support at a moment's notice, all day, every day – we can keep your instruments and your lab running at optimum levels. And eliminate the finger-pointing that can happen with piecemeal approaches.

Add to that the ability to deliver all the consumables and accessories you rely on, and it's clear: this is the one-stop shop you've always wanted to do business with.

The Titan MPS microwave sample preparation system. It's a microwave – and everything that comes with it.

THE ONE-STOP SHOP, REALIZED

With PerkinElmer, you benefit from the convenience of one provider for all your instrumentation, supplies, and support. For example:

- Our OneSource team has more than 400,000 multivendor instruments under its care, in 120 countries around the world
- The Titan MPS system is fully supported no matter where you are
- Your PerkinElmer support representative will be grounded in atomic spectroscopy and microwave digestion technologies and techniques, for end-to-end lab coverage
- All the consumables you need to keep your lab up and running are available from perkinelmer.com

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